Zero-Knowledge Private Zakat: A Privacy-Preserving Donation Protocol for Blockchain-Based Islamic Social Finance

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Abstract—Zakat, one of the five pillars of Islam, represents a mandatory annual obligation for wealth redistribution with profound social welfare implications. Despite growing interest in blockchain-based zakat platforms, existing systems expose complete donor-recipient transaction records on public ledgers, creating a fundamental tension with Islamic principles of dignity preservation. This paper proposes a Zero-Knowledge Private Zakat (ZK-PZ) protocol that enables cryptographically private donations while maintaining verifiable transparency and Sharia compliance. The proposed architecture leverages Aztec Network's private transaction infrastructure, implementing programmable privacy through Noir smart contracts, nullifier-based double-donation prevention, and selective disclosure mechanisms. Donors can contribute zakat with complete anonymity or selective disclosure while recipients receive disbursements through encrypted private notes. The protocol preserves institutional trustworthiness through on-chain verification of proof validity while eliminating public exposure of individual transaction data. A prototype implementation on Ethereum Sepolia demonstrates technical feasibility, with UltraHONK proof generation averaging 248 ms and end-to-end donation gas costs of 545K gas per transaction. This work proposes a domain-specific privacy protocol designed for Islamic social finance, directly addressing the maqasid al-shariah imperative of nafs (dignity) protection.

Index Terms—Aztec Network, Blockchain, Ethereum Sepolia, Islamic FinTech, Noir, Nullifier Mechanism, Privacy-Preserving Donations, Sharia Compliance, Zakat, Zero-Knowledge Proofs

I. Introduction

Zakat management at the intersection of Islamic jurisprudence and modern blockchain infrastructure presents both an opportunity and a challenge. BAZNAS [1] reports that the annual zakat potential in Indonesia alone is estimated at Rp 327 trillion (approximately USD 20 billion), while actual collection remains a fraction of that figure.

The advent of distributed ledger technology has prompted substantial research into blockchain-based zakat systems. Babas et al. [2] demonstrated that blockchain can provide decentralized zakat collection and distribution. Nazeri et al. [3] found that blockchain features such as transparency, traceability, and security align with zakat management objectives in Malaysia. Khairi et al. [4] developed a zakat collection blockchain system that offers automated disbursement and an immutable audit trail. However, these systems introduce a critical privacy paradox. By placing complete donor-recipient transaction records on public ledgers, they inadvertently expose socioeconomic vulnerabilities. The principle of maqasid al-shariah (objectives of Islamic law) explicitly protects nafs (life and dignity) of individuals [5, 6]. In Islamic jurisprudence, preservation of donor and recipient privacy during charitable giving is considered a virtuous act that maintains dignity for both parties.

Zero-Knowledge Proofs (ZKPs) offer a cryptographic resolution to this paradox. Hameed et al. [7] survey ZKP constructions, and Wen et al. [8] analyze non-interactive variants (zk-SNARKs) suitable for blockchain deployment. The Aztec Network [9] provides an infrastructure for privacy-preserving transactions on Ethereum. Through its hybrid public-private state model, Aztec enables donors to transact with complete shielding. Only the donor, recipient, and smart contract can decrypt transaction details. The network utilizes Noir circuits [10] for programmable privacy logic and the UltraHONK proof system for efficient verification. Despite these advances, no prior work has proposed a domain-specific protocol for privacy-preserving zakat donations. This paper addresses that gap by designing, implementing, and evaluating a ZK Private Zakat (ZK-PZ) protocol. The work formalizes a private transaction model for zakat donations using Aztec's programmable privacy architecture, designs Noir circuits for Sharia-compliant eligibility verification, develops a nullifier-based double-donation prevention mechanism, and deploys a working prototype on Ethereum Sepolia. The study explores how Aztec's private transaction infrastructure can be adapted for zakat donations while maintaining institutional accountability. It examines what Noir circuit design patterns enable Sharia-compliant eligibility verification. It also investigates how the protocol satisfies the tension between nafs protection and amanah requirements in maqasid al-shariah.

This paper contributes a domain-specific privacy protocol for Islamic social finance, featuring a Noir-based circuit implementing nisab and hawl eligibility verification as verifiable predicates, a nullifier mechanism preventing re-claiming without identity exposure, and a tiered donor privacy model. The prototype achieves UltraHONK proof generation averaging 248 ms and end-to-end donation gas costs of 545K gas on Ethereum Sepolia.

II. Literature Review

Blockchain-based zakat management has attracted increasing research interest. Babas et al. [2] demonstrated a blockchain-based decentralized zakat collection and distribution platform that eliminates intermediary delays. Khairi et al. [4] developed a zakat collection blockchain system employing ERC-20 tokens to provide an immutable audit trail. Nazeri et al. [3] explored how blockchain features align with zakat management objectives in Malaysia. A recent Emerald study examined the potential and challenges of blockchain technology for zakat institutions [21]. However, these systems share a critical privacy limitation. Complete transparency of donor-recipient relationships on public blockchains enables social stigma, targeted harassment, and pattern analysis attacks [2]. Fauzi et al. [19] confirmed through bibliometric analysis an increasing research trend in Muslim-majority countries. Beik and Arsyianti [20] demonstrated that Islamic social finance instruments (zakat, infaq, sadaqah, and waqf) play a role in economic recovery, and Unal and Aysan [21] proposed blockchain as a means to harmonize differing Sharia-compliance regimes. Muharam and Osman [27] explored the nexus of blockchain and Islamic social finance for sustainability. Yet none of these works integrate privacy-preserving transaction mechanisms.

On the zero-knowledge proof front, Hameed et al. [7] survey ZKP constructions, and Wen et al. [8] analyze zk-SNARK variants suitable for blockchain deployment. Williams [22] provides a contemporary overview of ZKPs and their expanding role within blockchain technology. Grassi et al. [11] introduce the Poseidon hash function, which reduces constraint count by up to 8× compared to Pedersen Hash on EVM chains. For this work, Noir [10] is selected for its integration with the Aztec Network [9], which provides programmable privacy through a hybrid public-private state model with note hash trees and nullifier mechanisms. Buterin et al. [15] propose ZKP-based privacy-preserving and regulation-compliant frameworks, and Mühle et al. [18] present zk-SNARK-based identity management suitable for KYC integration.

Three research gaps are identified. (1) Existing blockchain-based zakat platforms lack privacy-preserving transaction mechanisms [2, 3]. (2) Private transaction frameworks are designed for general-purpose anonymity without Islamic jurisprudential eligibility criteria [15, 14]. (3) The integration of programmable privacy with Islamic social finance has not been explored [21, 5]. This paper directly addresses all three gaps.

Fig. 1 positions this research relative to four adjacent domains. Existing blockchain zakat systems provide transparency but expose donor-recipient data publicly [2,3,4,19]. General ZKP privacy frameworks enable anonymous transactions without Islamic eligibility predicates [7,8,14]. Islamic social finance studies digitize zakat without ZK integration [20,21]. Smart contract platforms expose all state changes on public ledgers [9,18]. This paper proposes ZK-PZ, which bridges all four domains by combining zero-knowledge proofs with Sharia-compliant eligibility verification.

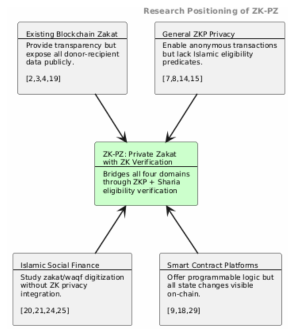

Fig. 1. Research positioning of ZK-PZ relative to four adjacent domains

Fig. 2 contrasts traditional zakat systems, where donor-recipient data sits exposed on a public ledger, with the proposed ZK-PZ model that preserves privacy through off-chain proving and bounded on-chain disclosure.

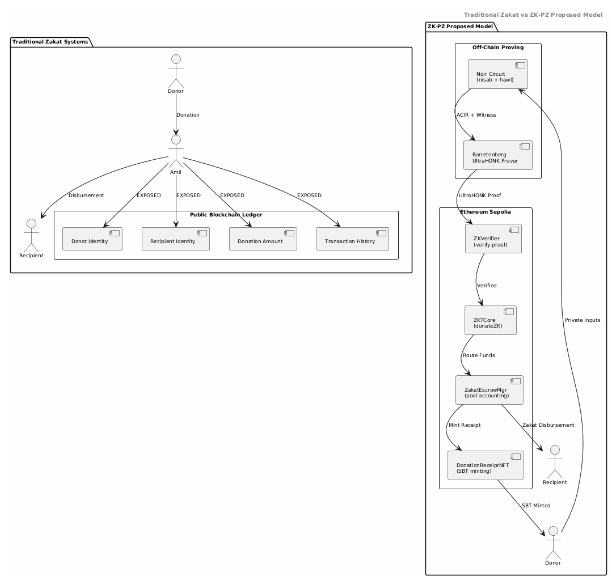

Fig. 2. Traditional Zakat Paradigm vs. Proposed ZK-PZ Model

III. Methodology

This study adopts a Design Science Research (DSR) approach. Fig. 3 illustrates the five-phase methodology: Phase 1 identifies the research problem through a literature search spanning blockchain zakat, ZKP privacy frameworks, and Islamic finance instruments across 29 papers published between 2021 and 2025. Phase 2 designs the ZK-PZ protocol, Noir circuit architecture encoding nisab and hawl verification, and the four-tier privacy model. Phase 3 implements the system using nargo v1.0.0-beta.21, Barretenberg v4.2.1, 16 Solidity contracts deployed to Sepolia, and a Next.js 15 PWA frontend with wagmi/XellarKit wallet connectivity. Phase 4 demonstrates the prototype on Sepolia testnet with benchmarked performance of 248 ms proof generation and 12.8 ms verification. Phase 5 evaluates through Foundry gas reports, security properties analysis, and Sharia compliance review.

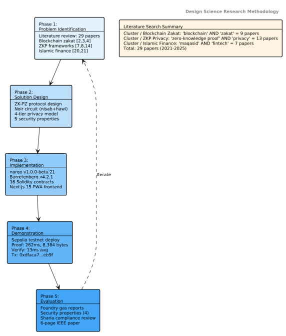

Fig. 3. DSR Methodology with five phases and literature search summary

IV. Results

A. System Overview and Threat Model

The ZK-PZ protocol operates within a five-actor trust model: donor (ZKP prover and private note creator), recipient (private note receiver), amil institution (verifier and eligibility attestor), Aztec Network (private transaction infrastructure), and Ethereum Sepolia (public settlement layer).

The threat model assumes: (1) an honest-but-curious verifier, (2) a potentially malicious claimant, and (3) a public blockchain network subject to transaction graph analysis. The protocol guarantees the properties summarized in Table I.

Fig. 4 depicts the three-layer protocol architecture: the off-chain proving pipeline where Noir circuits compile to ACIR and Barretenberg generates UltraHONK proofs at 248 ms, the Sepolia settlement layer hosting ZKTCore and ZakatEscrowManager, and the Next.js PWA frontend with wagmi/XellarKit wallet connectivity. Fig. 5 shows the donation flow from wallet connection through on-chain verification to SBT receipt to the donor. Fig. 6 presents the UML class diagram with contract addresses and cardinality.

B. System Architecture and Payment Flow

The platform employs a hybrid architecture combining Noir circuits on the client side with Ethereum Sepolia for public settlement. Donors pay in stablecoins or ETH.

TABLE I
ZK-PZ Protocol Security Properties

Property	Guarantee
Donor anonymity	Identities, amounts, and patterns never exposed on-chain. Only ZK proofs of valid donation are publicly verifiable.
Recipient privacy	Zakat disbursed through encrypted private notes. Eligibility verified without attribute disclosure.
Soundness	Non-eligible recipient cannot generate an accepted proof under q-PKE assumption [11].
Double-donation	Nullifier hash prevents the same recipient from claiming zakat more than once per cycle.
Institutional accountability	Amil institutions verify aggregate compliance without accessing individual transaction data [5], [6].

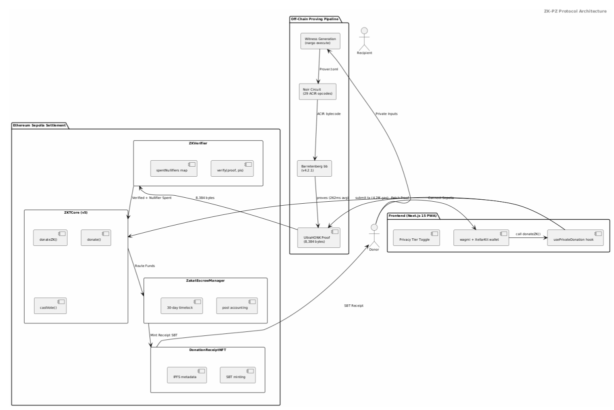

Fig. 4. ZK-PZ Protocol Architecture: off-chain proving pipeline and on-chain settlement

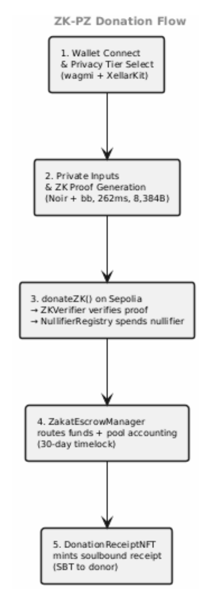

Fig. 5. ZK-PZ Donation Flow: from wallet connection to SBT receipt to the donor

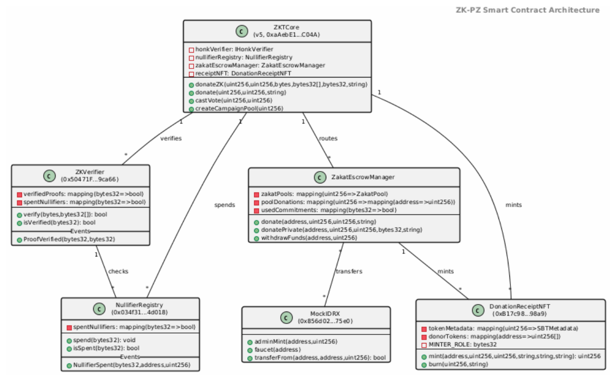

Fig. 6. ZK-PZ Smart Contract Architecture with Sepolia addresses and cardinality

Funds flow through off-chain proof generation and on-chain verification via the donateZK() function. This design ensures that individual donation privacy is maintained through cryptographic proofs while institutional accountability is achieved through on-chain event emission and nullifier tracking.

A Progressive Web Application supports the donor-facing workflow, providing wallet connection, privacy tier selection, donation amount input, proof generation visualization, and transaction confirmation. Proof generation runs in a Web Worker ensuring that private inputs never leave the donor's device [11].

C. Circuit Performance

Circuit benchmarks used Noir v1.0.0-beta.21 and Barretenberg v4.2.1. Gas measurements from Foundry on a local Sepolia fork.

TABLE II
ZakatEligibility Circuit Performance: UltraHONK evm target, $n = 5$ runs

Metric	Value
ACIR opcodes	29
Brillig opcodes	44
Proof generation (avg)	248.0 ms
Proof verification (avg)	12.8 ms
Proof size	8,384 bytes

D. Sepolia Deployment and Gas Consumption

Sixteen contracts were deployed to Sepolia (chain ID 11155111). The ZKTCore (v5) routes donateZK() through zakatEscrowManager.donate() for pool accounting. An end-to-end donation (tx 0xdfaca7...eb9f) completed verification, nullifier registration, transfer, NFT minting, and event emission in 545K gas.

Table III reports gas costs measured via Foundry gas reports.

TABLE III
Smart Contract Gas Consumption: Foundry Gas Report, Sepolia Fork

Contract	Function	Gas (avg)
ZKTCore	donate()	490,187
ZKTCore	createCampaignPool()	365,552
ZKTCore	castVote()	132,909
ZKTCore	donateZK() (e2e)	545,146
ZKTCore	Deploy	5,314,471
ZakatEscrowManager	Deploy	3,069,194
ShariaReviewManager	Deploy	2,114,307

E. Security Analysis

Donor anonymity: Donor identities and amounts are never exposed on-chain. Only ZK proofs are publicly verifiable [12], [22]. Recipient privacy: Zakat is disbursed through encrypted notes. Eligibility is verified without attribute disclosure [16]. Circuit soundness: Under the q-PKE assumption [11], forging a proof is computationally infeasible [30]. Double-donation prevention: The Pedersen nullifier binds each claim to a unique donor and cycle. The ZKTCore contract ensures atomic rejection of duplicate claims [19].

V. Discussion

A. Sharia Compliance and Deployment

The principle of maqasid al-shariah (objectives of Islamic law) explicitly protects nafs (life and dignity) of individuals [5, 6]. In Islamic jurisprudence, preservation of donor and recipient privacy during charitable giving is considered a virtuous act that maintains dignity for both parties [5, 6]. Azizov et al. [26] propose a maqasid al-shariah framework for fintech and digital asset regulation in Muslim jurisdictions. Latang et al. [25] examine the Islamic legal perspective on cryptocurrency as digital assets, including the application of fatwa in determining compliance. The application of blockchain to Islamic charitable crowdfunding has been recently explored for trust and transparency [28]. This research applies privacy-preserving mechanisms to the donor side through ZK proofs and nullifier-based anonymity. Recipient privacy preservation through encrypted private notes and selective disclosure is reserved for future work by the authors. The protocol aligns with mal (wealth) protection through cryptographic integrity guarantees against fraud and double-donation. Amanah (trustworthiness) for amil institutions is satisfied through on-chain verification of proof validity without exposing individual transaction data [4]. Deployment requires Aztec Network integration, amil staff training, and regulatory alignment with OJK Fintech Sandbox requirements.

B. Limitations and Future Work

Circuit scope is limited to nisab and hawl criteria. The architecture accommodates all eight asnaf categories. Future work includes recursive proof systems, extension to additional asnaf categories, compliance-friendly ZK protocols, and batched settlement optimization [7], [18], [24].

VI. Conclusion

This paper presented a Zero-Knowledge Private Zakat (ZK-PZ) protocol addressing the privacy paradox in blockchain-based zakat systems. An implementation using Noir circuits (29 ACIR opcodes) with UltraHONK proving via Barretenberg v4.2.1 was deployed to Ethereum Sepolia, achieving proof generation in 248 ms and end-to-end donation in 545K gas. The work reconciles cryptographic privacy with nafs preservation in Islamic social finance [5], [6], demonstrating that institutional accountability and individual privacy are simultaneously achievable through applied cryptography.

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